Lab 7 Milestones 1 and 2 - results

1. WALL-CLOCK TIME
   First 1000 consecutive fast readings: 1.2096 s (1.21 ms per reading)
   (the saved run has 2048 consecutive readings; all are used for the averaging plot)

2. AVERAGING LAW (std of the average of N readings, non-overlapping groups)

N	groups	measured std [C]	1/sqrt(N) line [C]
1	2048	0.016886	0.016886
2	1024	0.016853	0.011940
4	512	0.016843	0.008443
8	256	0.016833	0.005970
16	128	0.016742	0.004221
32	64	0.016568	0.002985
64	32	0.016169	0.002111
128	16	0.015286	0.001493
256	8	0.014323	0.001055

3. DEPARTURE N
   N = 4 (measured 0.016843 C vs line 0.008443 C)
   Rule: first N where the measured std is more than 1.5x the sigma_1/sqrt(N) line.

4. 300 ms COMPARISON (100 readings each)
   First 100 fast readings: std = 0.000000 C
   100 readings ~300 ms apart: std = 0.017963 C
   The fast readings repeat one sensor conversion; the 300 ms readings are independent.

16-BIT MODE: CONFIRMED
   lab7_m1.py writes config register 0x03 = 0x80 and aborts unless it reads back 0x80;
   2142 of 2148 saved raw codes use the 1/128 C steps that only 16-bit mode gives.

M1a (5 power levels, 180 s settling wait per level, sample std ddof=1)

level	power [W]	mean T [C]	std T [C]	n
1	0.0000	29.6909	0.01524	100
2	0.0690	30.9631	0.01962	100
3	0.1411	32.2677	0.02294	100
4	0.2118	33.6319	0.02614	100
5	0.2827	34.7827	0.02121	100

   power = measured |V x I| from the supply, not the target

M2a (34461A DC volts across the unpowered resistor, range 0.1 V, autozero ON)

NPLC	integration [ms]	n	mean [uV]	std [uV]
0.02	0.333	200	426.3	200.3
0.2	3.333	200	401.8	149.1
1	16.667	200	456.7	26.3
10	166.667	200	463.4	23.1
100	1666.667	200	461.7	25.9

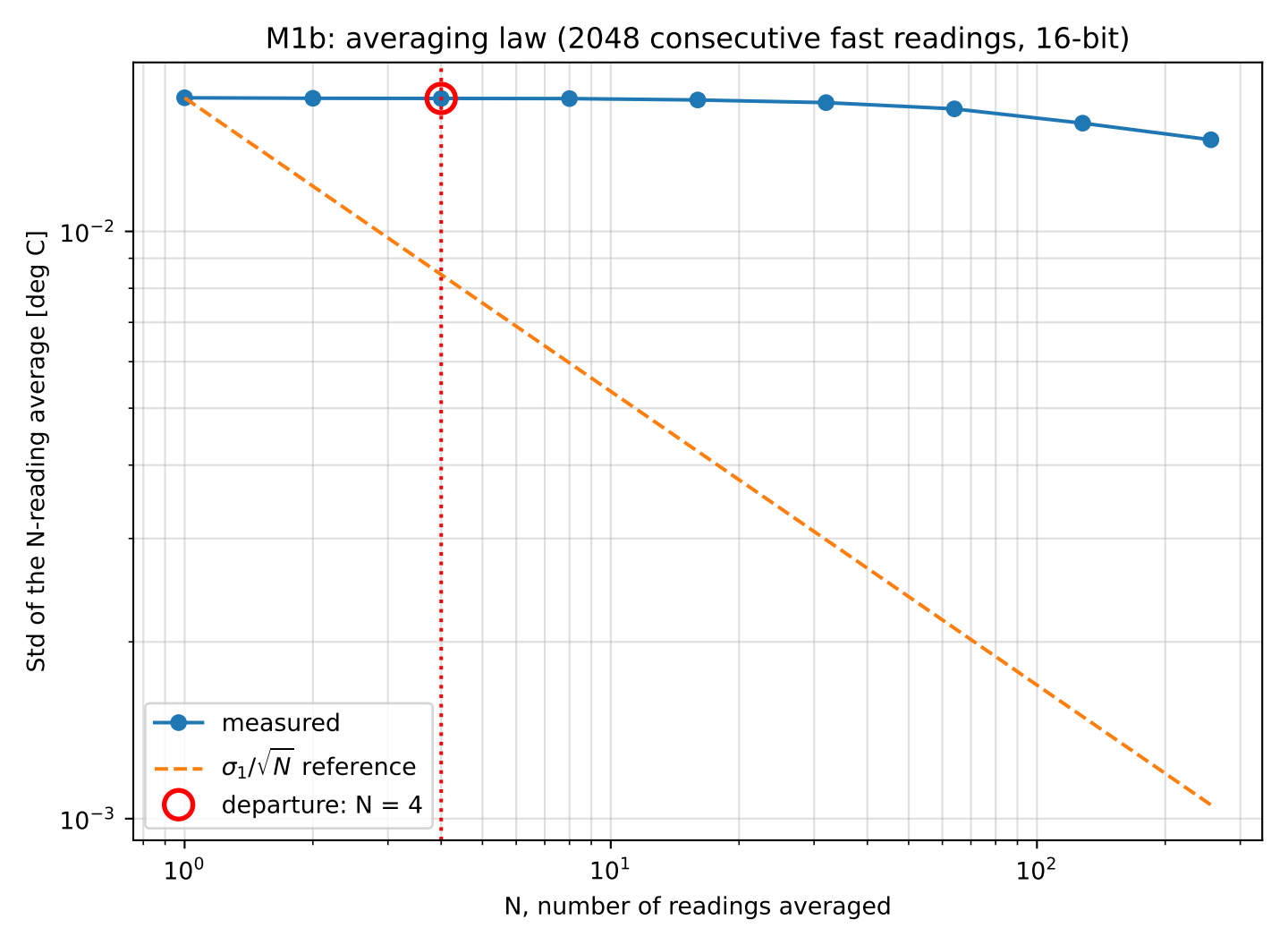
   1 PLC = 16.7 ms integrates exactly one 60 Hz cycle, so the pickup averages to zero

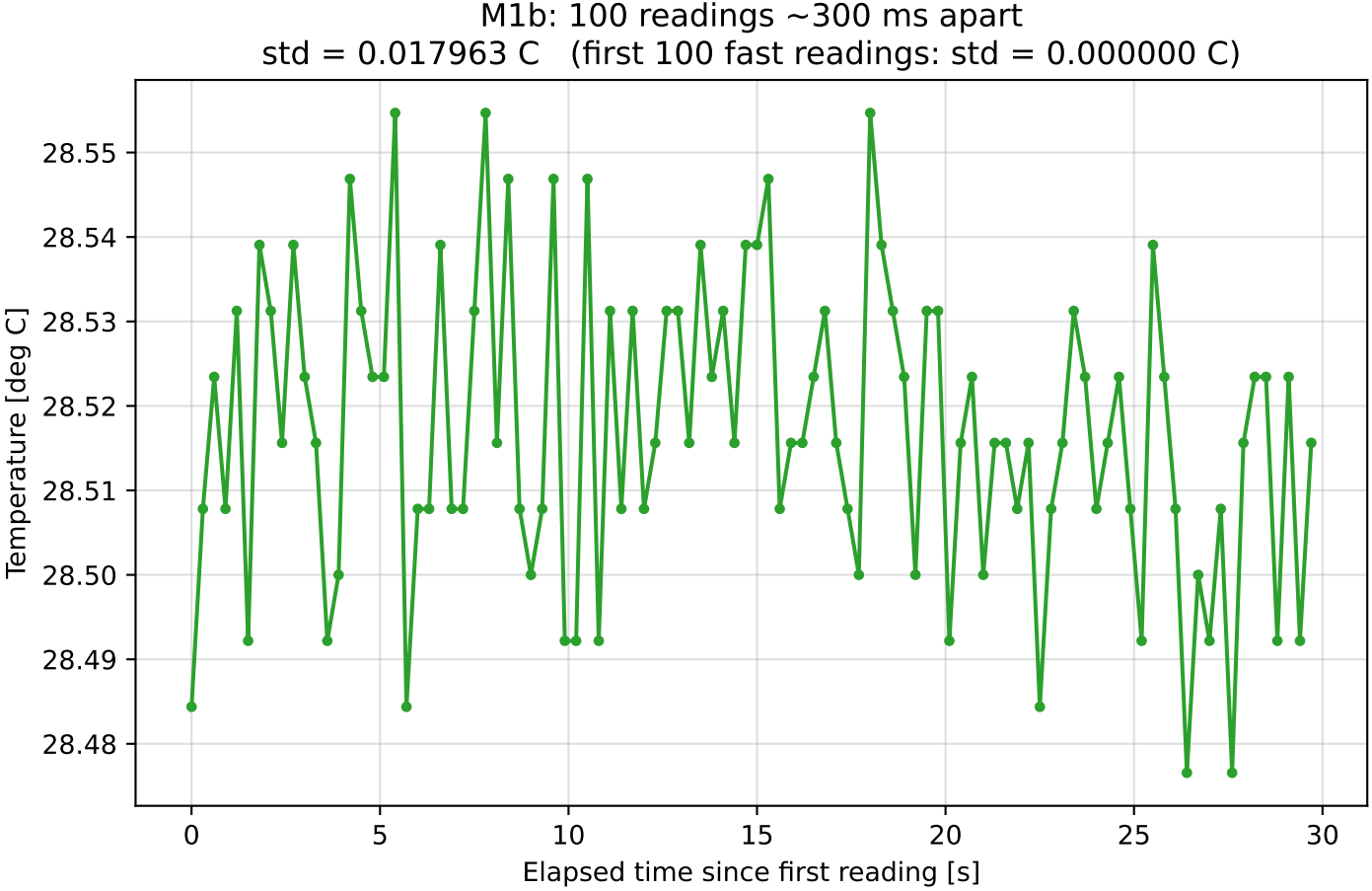
M2b (scope FFT of the same resistor node, 1 s records, Hann window)

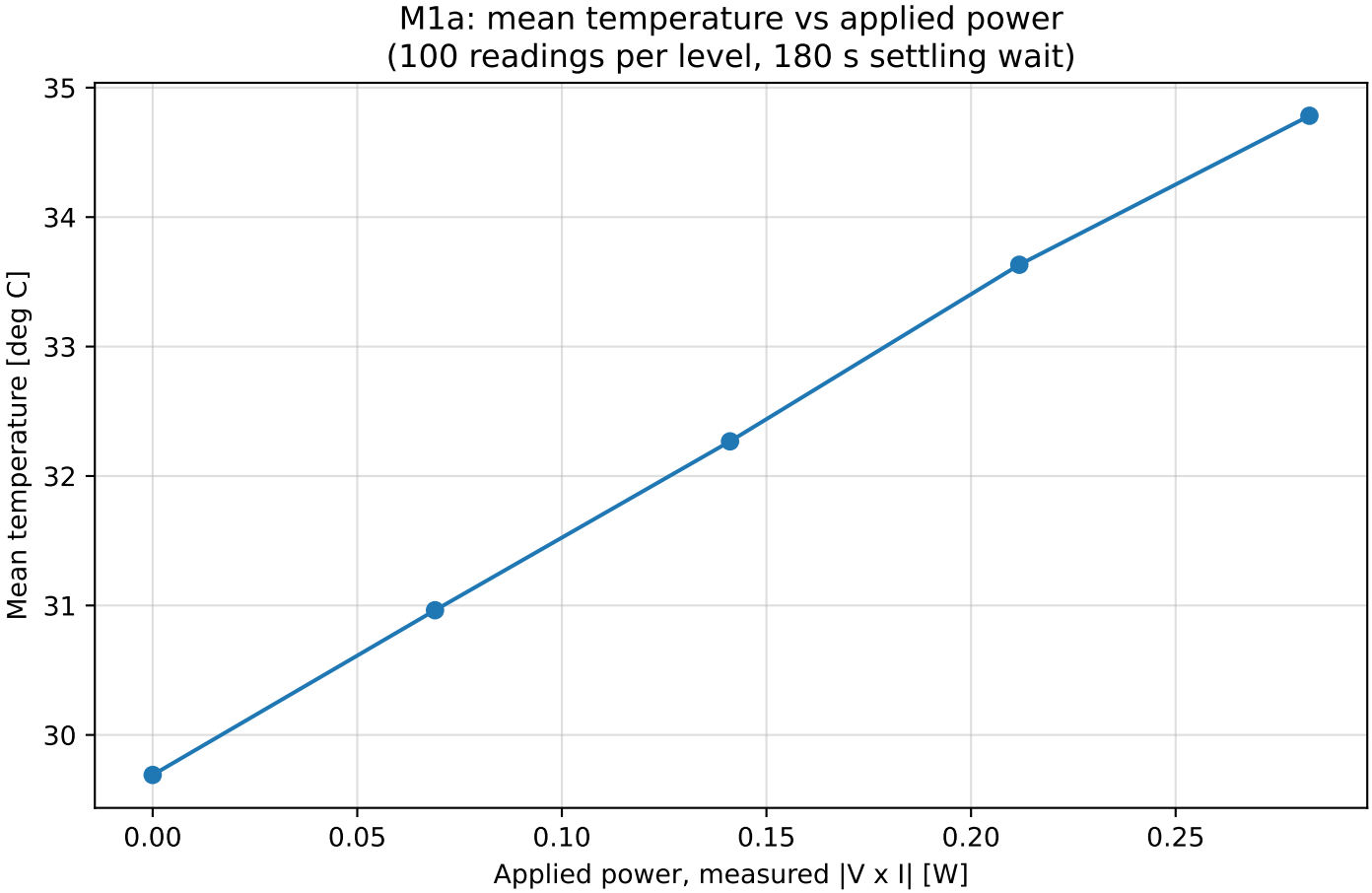
loop	60 Hz amplitude [V peak]	noise floor [V]	peak / floor
short	2.070e-04 at 60.0 Hz	1.62e-06	127x
long	4.813e-03 at 60.0 Hz	3.19e-06	1511x

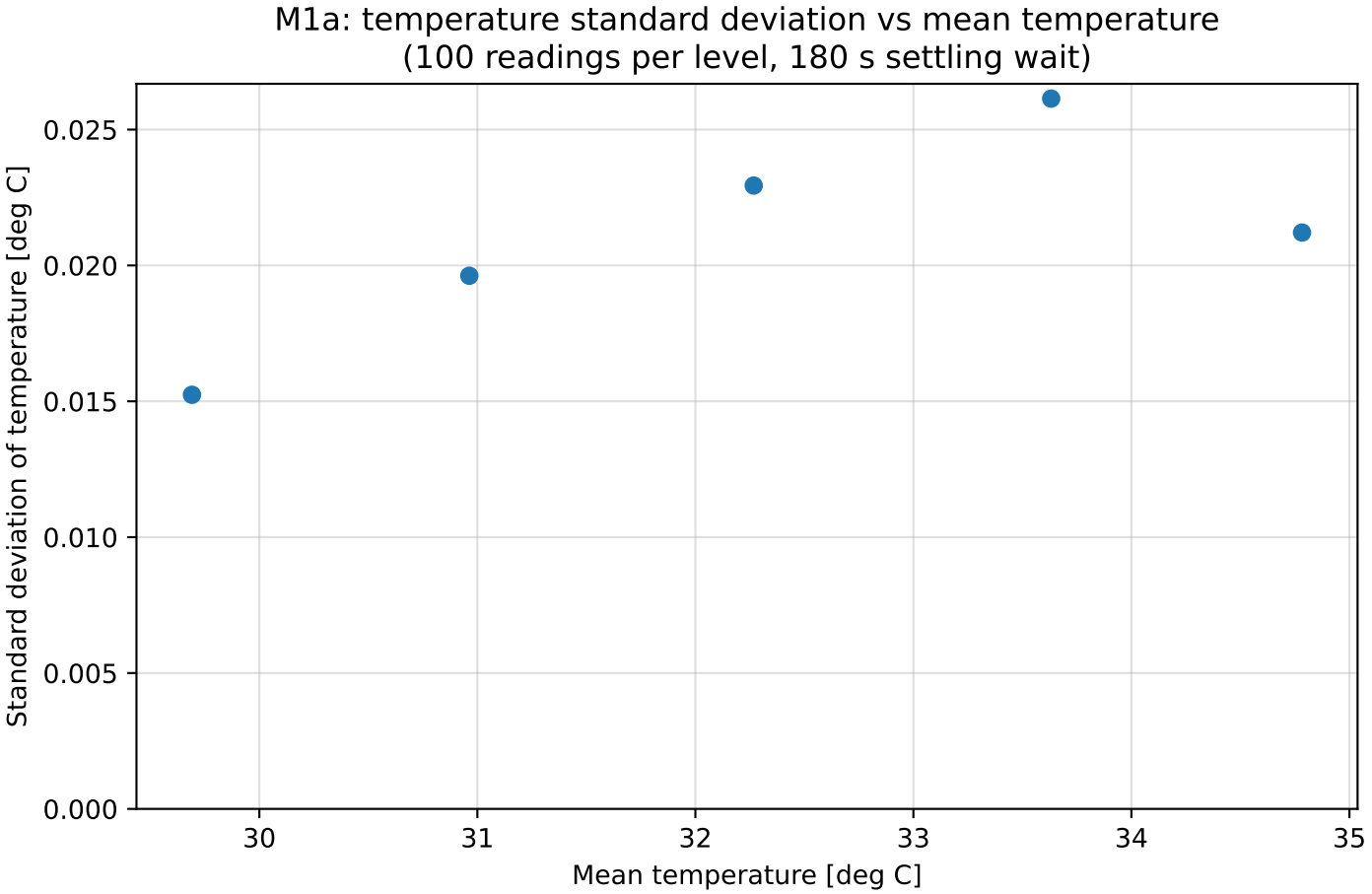
   amplitude ratio long / short = 23.3
   loop areas 1 cm^2 (spring) and 900 cm^2 (long wire): area ratio 900
   60 Hz + odd harmonics = interference: fixed by a smaller loop, not by averaging

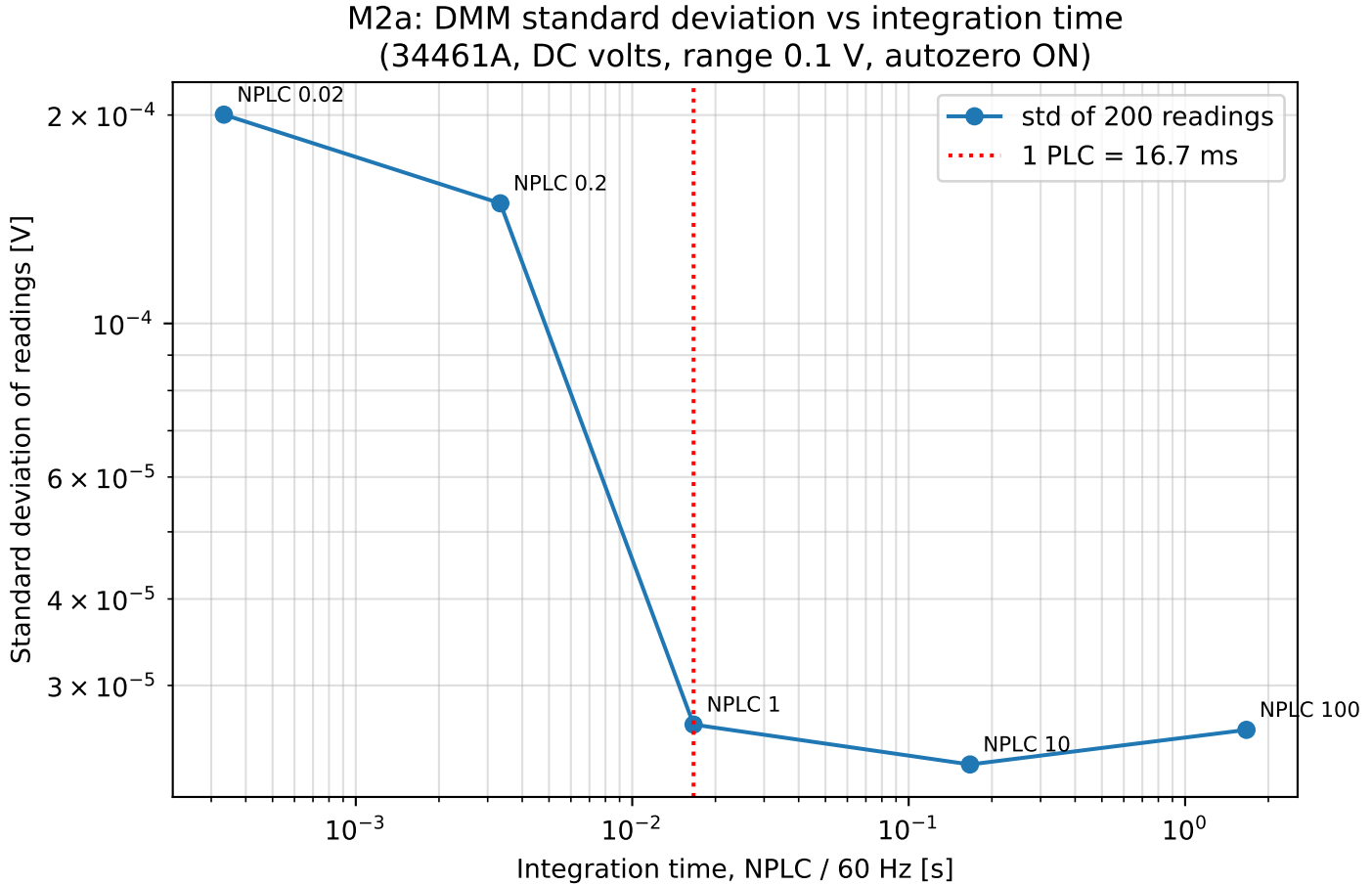
M2c (SDA_0 through the read-address ACK, 10 captures, triggered on the PMOD marker)
   10-90 % rise: 924.8 +/- 11.3 ns   (10 kohm pull-up charging the bus)
   90-10 % fall: 8.32 +/- 0.05 ns   (sensor's open-drain transistor pulling low)
   rise / fall = 111; tau = t_rise / ln 9 = 421 ns (fit 431 ns)
   bus capacitance C = tau / R = 42.1 pF (R = 10 kohm, includes the probe)
   standard mode, 100 kHz max rise 1000 ns: OK
   fast mode, 400 kHz   max rise 300 ns: TOO SLOW

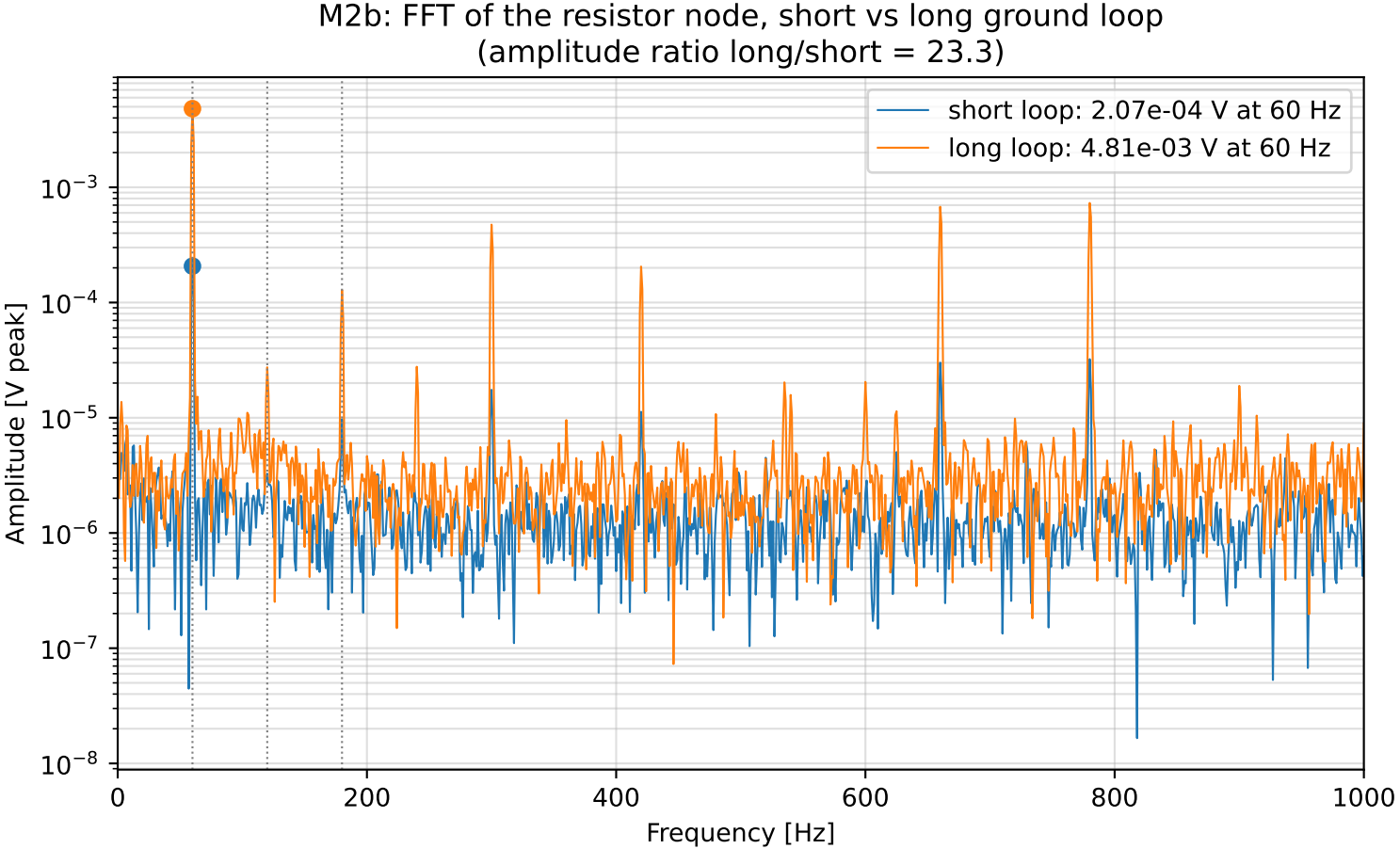

M1b: averaging law (2048 consecutive fast readings, 16-bit)
Std of the N-reading average [deg C]
10^{-2}
10^{-3}
10^{0}
10^{1}
10^{2}
N, number of readings averaged
measured
$\sigma_1/\sqrt{N}$ reference
departure: N = 4

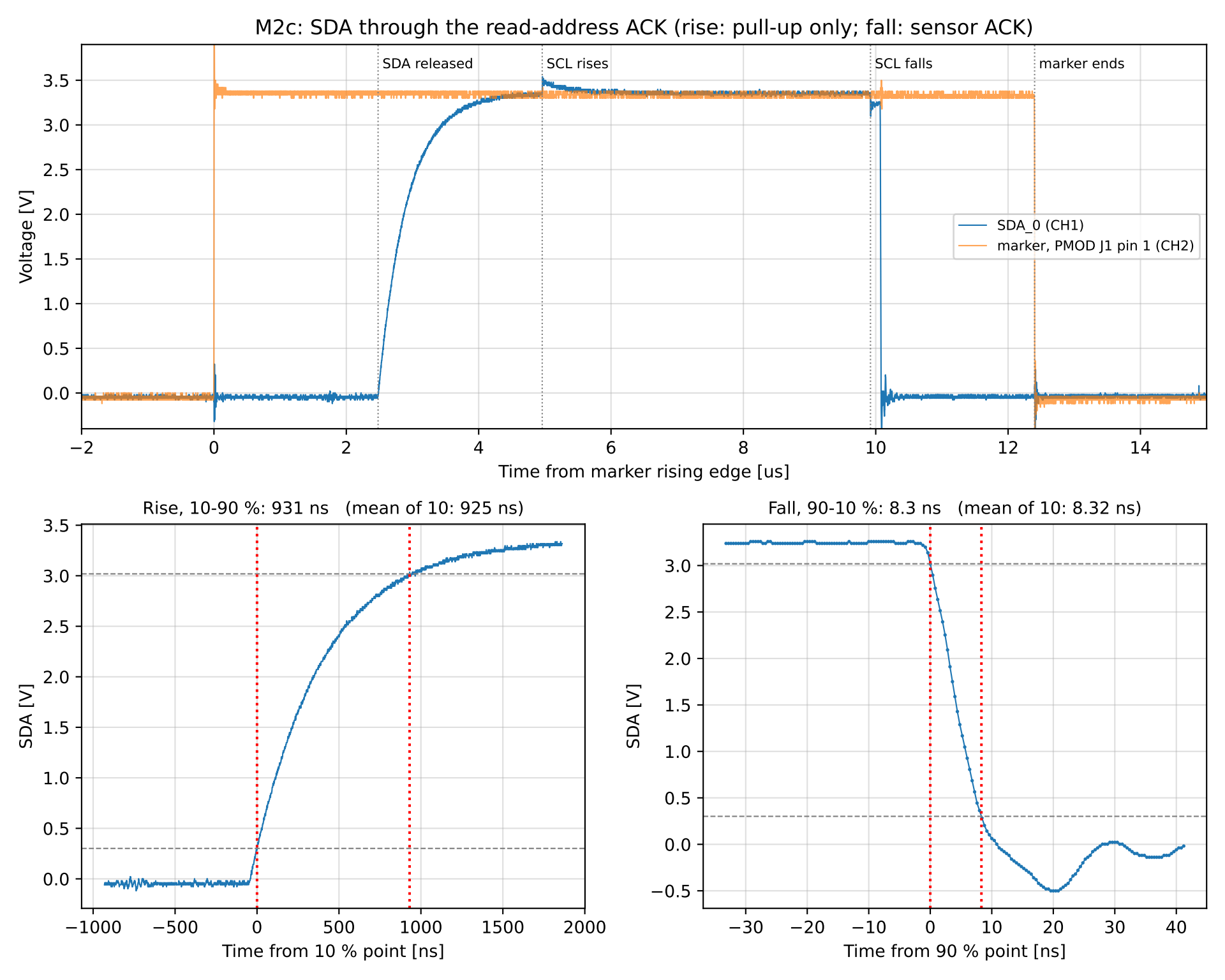

M1b: 100 readings ~300 ms apart
std = 0.017963 C (first 100 fast readings: std = 0.000000 C)
Temperature [deg C]
28.55
28.54
28.53
28.52
28.51
28.50
28.49
28.48
0
5
10
15
20
25
30
Elapsed time since first reading [s]


M1a: mean temperature vs applied power
(100 readings per level, 180 s settling wait)
Mean temperature [deg C]
35
34
33
32
31
30
0.00
0.05
0.10
0.15
0.20
0.25
Applied power, measured |V x I| [W]


M1a: temperature standard deviation vs mean temperature
(100 readings per level, 180 s settling wait)
Standard deviation of temperature [deg C]
0.000
0.005
0.010
0.015
0.020
0.025
30
31
32
33
34
35
Mean temperature [deg C]


M2a: DMM standard deviation vs integration time
(34461A, DC volts, range 0.1 V, autozero ON)
std of 200 readings
1 PLC = 16.7 ms
NPLC 0.02
NPLC 0.2
NPLC 1
NPLC 10
NPLC 100
Standard deviation of readings [V]
Integration time, NPLC / 60 Hz [s]


M2b: FFT of the resistor node, short vs long ground loop
(amplitude ratio long/short = 23.3)
short loop: 2.07e-04 V at 60 Hz
long loop: 4.81e-03 V at 60 Hz
Amplitude [V peak]
Frequency [Hz]
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-8}
0
200
400
600
800
1000


M2c: SDA through the read-address ACK (rise: pull-up only; fall: sensor ACK)
SDA released
SCL rises
SCL falls
marker ends
SDA_0 (CH1)
marker, PMOD J1 pin 1 (CH2)
Voltage [V]
Time from marker rising edge [us]
Rise, 10-90 %: 931 ns (mean of 10: 925 ns)
SDA [V]
Time from 10 % point [ns]
Fall, 90-10 %: 8.3 ns (mean of 10: 8.32 ns)
SDA [V]
Time from 90 % point [ns]